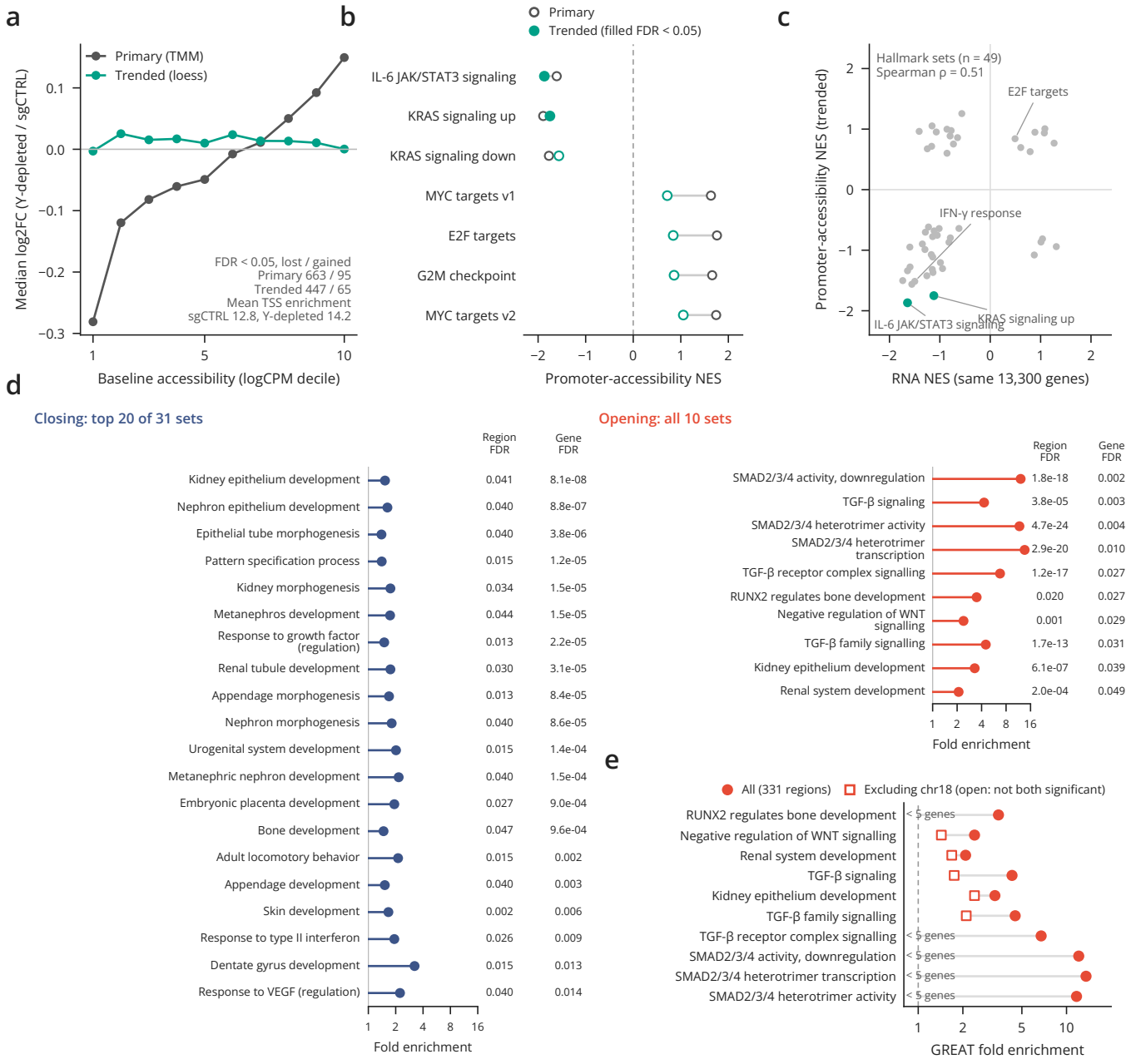
a
Primary (TMM)
Trended (loess)
Median log2FC (Y-depleted / sgCTRL)
FDR < 0.05, lost / gained
Primary 663 / 95
Trended 447 / 65
Mean TSS enrichment
sgCTRL 12.8, Y-depleted 14.2
Baseline accessibility (logCPM decile)
b
Primary
Trended (filled FDR < 0.05)
IL-6 JAK/STAT3 signaling
KRAS signaling up
KRAS signaling down
MYC targets v1
E2F targets
G2M checkpoint
MYC targets v2
Promoter-accessibility NES
c
Hallmark sets (n = 49)
Spearman ρ = 0.51
E2F targets
IFN-γ response
KRAS signaling up
IL-6 JAK/STAT3 signaling
Promoter-accessibility NES (trended)
RNA NES (same 13,300 genes)
d
Closing: top 20 of 31 sets
Region FDR
Gene FDR
Kidney epithelium development 0.041 8.1e-08
Nephron epithelium development 0.040 8.8e-07
Epithelial tube morphogenesis 0.040 3.8e-06
Pattern specification process 0.015 1.2e-05
Kidney morphogenesis 0.034 1.5e-05
Metanephros development 0.044 1.5e-05
Response to growth factor (regulation) 0.013 2.2e-05
Renal tubule development 0.030 3.1e-05
Appendage morphogenesis 0.013 8.4e-05
Nephron morphogenesis 0.040 8.6e-05
Urogenital system development 0.015 1.4e-04
Metanephric nephron development 0.040 1.5e-04
Embryonic placenta development 0.027 9.0e-04
Bone development 0.047 9.6e-04
Adult locomotory behavior 0.015 0.002
Appendage development 0.040 0.003
Skin development 0.002 0.006
Response to type II interferon 0.026 0.009
Dentate gyrus development 0.015 0.013
Response to VEGF (regulation) 0.040 0.014
Fold enrichment
Opening: all 10 sets
Region FDR
Gene FDR
SMAD2/3/4 activity, downregulation 1.8e-18 0.002
TGF-β signaling 3.8e-05 0.003
SMAD2/3/4 heterotrimer activity 4.7e-24 0.004
SMAD2/3/4 heterotrimer transcription 2.9e-20 0.010
TGF-β receptor complex signalling 1.2e-17 0.027
RUNX2 regulates bone development 0.020 0.027
Negative regulation of WNT signalling 0.001 0.029
TGF-β family signalling 1.7e-13 0.031
Kidney epithelium development 6.1e-07 0.039
Renal system development 2.0e-04 0.049
Fold enrichment
e
All (331 regions)
Excluding chr18 (open: not both significant)
RUNX2 regulates bone development
Negative regulation of WNT signalling
Renal system development
TGF-β signaling
Kidney epithelium development
TGF-β family signalling
TGF-β receptor complex signalling
SMAD2/3/4 activity, downregulation
SMAD2/3/4 heterotrimer transcription
SMAD2/3/4 heterotrimer activity
< 5 genes
GREAT fold enrichment